

Ideation Phase: Define the Problem Statements

Date	2 July 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	3 Marks

Customer Problem Statement Template:

Introduction: Floods are among the most devastating natural disasters, causing significant loss of life, property, and infrastructure every year. Conventional flood forecasting methods often rely on manual analysis and delayed reporting, making timely decision-making difficult during emergencies.

Solution Approach: The *Rising Waters* system leverages Machine Learning to analyze historical weather and rainfall data, including annual rainfall, cloud cover, seasonal rainfall, and other meteorological parameters, to predict the likelihood of flood events. By providing accurate and timely flood risk predictions, the system enables disaster management authorities to plan early evacuations, allocate emergency resources efficiently, and improve disaster preparedness.

Problem Statement Framework Diagram:



Problem Statement Table:

S.No	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A disaster management officer	Predict flood risk early and plan timely evacuation and resource allocation.	Existing flood prediction methods are often delayed and require manual analysis.	Weather conditions change rapidly, making it difficult to assess flood risk accurately in time.	Pressured and underprepared during emergency situations.